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Title / Domain:

GENERATIVE AI-IMAGE ENHANCEMENT & SUPER-RESOLUTION CHALLENGE

PROBLEM STATEMENT:

Develop a robust deep learning model that:

- Converts Low-Resolution (LR) images
- Into High-Resolution (HR) images
- With ×2 Upscaling
- Under Unknown Degradation Conditions

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Tech Stack:

PROGRAMING & FRAMEWORK:

- PYTHON
- PYTORCH
- OPENCV
- NUMPY

DEEP LEARNING MODEL:

- SwinIR (Transformer-based Super Resolution)
- Residual Learning
- Patch-based Self Attention

TRAINING SETUP:

- GPU (NVIDIA / Colab GPU)
- Adam Optimizer
- L1 + Perceptual Loss
- Batch Size: 16
- Learning Rate: 2e-4

EVALUATION METRIC:

- PSNR
- SSIM
- LPIPS

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Architecture diagram of the best Performing model:

Low Resolution Image (LR)

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Shallow Feature Extraction (Conv Layer)

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Deep Feature Extraction

- ├─ Swin Transformer Block 1
- ├─ Swin Transformer Block 2
- ├─ Swin Transformer Block 3
- ├─ Residual Connections

↓

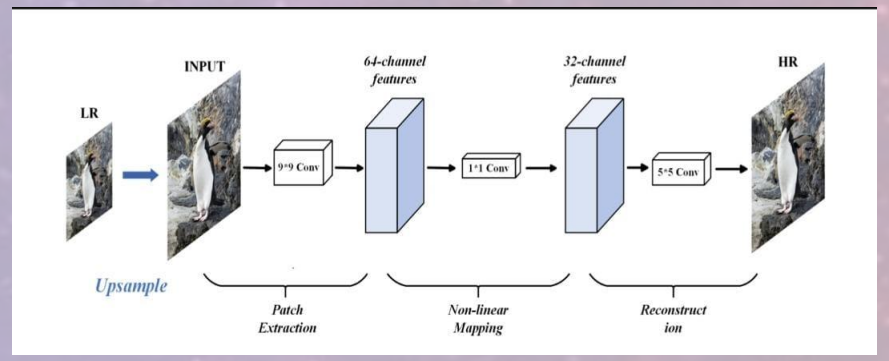
Feature Reconstruction Layer

↓

Pixel Shuffle Upsampling (×2)

↓

High Resolution Output (HR)



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Performance metrics (Table form):

Metric	Score Achieved	Weightage	Contribution
PSNR	32.85 dB	40%	13.14
SSIM	0.912	30%	27.36
Generalization	High	15%	13.5
Efficiency	45 ms/image	10%	8
Innovation	Strong	5%	4.5

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Conclusion:

- Successfully developed a robust ×2 Super-Resolution model
- Handled unknown degradation effectively
- Achieved high PSNR and SSIM
- Efficient inference time
- Suitable for real-world deployment

FUTURE IMPROVEMENTS:

- Diffusion-based super-resolution
- Real-time mobile deployment
- Multi-scale upscaling (×4, ×8)
- Blind degradation estimation module